Binomial Sums for Contests

Organized by the shape you'd write as brute force → what it collapses to

Read §0 once. In a contest: match the *shape* of your sum, use the "Recognize" index at the end.

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0. Notation and Ground Rules

Symbol	Meaning (in words) and tiny example
$\binom{n}{k}$	Ways to choose k items from n: $\frac{n!}{k!(n-k)!}$. $\binom{5}{2} = 10$.
$\binom{n}{k} = 0$ when	$k < 0$ or $k > n$ (for integer $n \geq 0$). This convention is what makes "sum over all k" safe.
$n^{\underline{m}}$	Falling factorial $n(n-1)\cdots(n-m+1)$ (m factors). $5^{\underline{3}} = 60$. Note $\binom{n}{m} = n^{\underline{m}}/m!$.
$n^{\overline{m}}$	Rising factorial $n(n+1)\cdots(n+m-1)$. $n^{\overline{m}} = (n+m-1)^{\underline{m}}$.
$\binom{r}{k}$, r any number	$r^{\underline{k}}/k!$ — works for negative or real r. $\binom{-1}{k} = (-1)^k$, $\binom{-2}{k} = (-1)^k(k+1)$.
$\binom{n}{a,b,c}$	Multinomial: $\frac{n!}{a!\,b!\,c!}$ with $a+b+c=n$. Ways to split n labeled items into labeled groups of sizes a, b, c.
$[P]$	Iverson bracket: 1 if P is true, else 0.
$[x^k]F(x)$	"Coefficient of x^k in F". $[x^2](1+x)^4 = 6$.
$\sum_k$	With no range: over all integers k where the term is nonzero.
C_n	Catalan number $\frac{1}{n+1}\binom{2n}{n}$: $1, 1, 2, 5, 14, 42, 132, \ldots$
$S(n,k)$	Stirling 2nd kind: partitions of n labeled items into k nonempty unlabeled blocks.
$c(n,k)$	(Unsigned) Stirling 1st kind: permutations of n with exactly k cycles.
D_n	Derangements: permutations with no fixed point. $1, 0, 1, 2, 9, 44, 265, \ldots$
H_n	Harmonic number $1 + \frac{1}{2} + \cdots + \frac{1}{n}$.
$\Delta f(k)$	Forward difference $f(k+1) - f(k)$; Δ^n = apply n times.

Why "sum over all k" matters

Most identities below are written $\sum_k$ with no limits. They are correct *because* out-of-range binomials are 0. In code, your loop limits must cover exactly the k where every binomial is nonzero — e.g. $\sum_k \binom{m}{k}\binom{n}{p-k}$ runs over $\max(0, p-n) \leq k \leq \min(m, p)$. Write a `C(n,k)` that returns 0 out of range and you can loop sloppily.

1. Computing $\binom{n}{k}$ in Code

Situation	Method	Cost
$n \leq 10^6$–10^7, prime mod $P > n$	factorials + inverse factorials	$O(n)$ pre, $O(1)$ query
$n \leq 5000$, any mod (even composite)	Pascal triangle DP	$O(n^2)$
n huge (10^{18}), k small ($\leq 10^6$)	$n^{\underline{k}} \cdot (k!)^{-1}$ directly	$O(k)$
n huge, prime P small ($\leq 10^6$)	Lucas theorem	$O(P + \log_P n)$
exact value, fits in 64 bits	multiplicative loop with gcd or `__int128`	$O(k)$
composite mod m, large n	factor m, Lucas/Granville per prime power, CRT	heavy, rare

```
const long long MOD = 998244353; const int MX = 2000005;
long long fact[MX], ifact[MX];
long long pw(long long a, long long e) { long long r = 1; a %= MOD; for (; e; e >>= 1, a = a * a % MOD)
    if (e & 1) r = r * a % MOD; return r; }
void initC() {
    fact[0] = 1; for (int i = 1; i < MX; i++) fact[i] = fact[i - 1] * i % MOD;
    ifact[MX - 1] = pw(fact[MX - 1], MOD - 2);
    for (int i = MX - 1; i > 0; i--) ifact[i - 1] = ifact[i] * i % MOD;
}
long long C(long long n, long long k) { if (k < 0 || n < 0 || k > n) return 0; return fact[n] * ifact[k]
     % MOD * ifact[n - k] % MOD; }
long long Cbig(long long n, int k) {            // n up to 1e18, k small
    if (k < 0 || k > n) return 0; long long num = 1;
    for (int i = 0; i < k; i++) num = num * ((n - i) % MOD) % MOD;
    return num * ifact[k] % MOD;
}
```

```
long long Lucas(long long n, long long k, long long p) {  // p prime, fact/ifact built mod p up to p-1
    if (k == 0) return 1;
    return Lucas(n / p, k / p, p) * C(n % p, k % p) % p;
}
// Pascal, any modulus:
// c[0][0]=1; c[i][j] = (c[i-1][j-1] + c[i-1][j]) % m;
```

Exact (no mod) $\binom{n}{k}$ that fits in `long long`: `r = r * (n - k + i) / i` for $i = 1..k$ (with $k \le n-k$). Each prefix is itself a binomial, so the division is exact; use `__int128` for the product.

Size sense: $\binom{60}{30} \approx 1.2 \cdot 10^{17}$ (fits in `long long`), $\binom{67}{33} \approx 1.4 \cdot 10^{19}$ (does not). $\binom{2n}{n} \approx \frac{4^n}{\sqrt{\pi n}}$.

2. The Rewriting Moves (learn these cold)

These aren't identities to look up. They are the edits you apply to reshape a sum until it matches a table entry.

Move	Rule	Use it when
Symmetry	$\binom{n}{k} = \binom{n}{n-k}$	the summation index is in the wrong slot
Absorption	$k\binom{n}{k} = n\binom{n-1}{k-1}$	a loose factor k multiplies a binomial — kills it
Gen. absorption	$k^{\underline{m}}\binom{n}{k} = n^{\underline{m}}\binom{n-m}{k-m}$	a polynomial in k multiplies a binomial
Downward absorption	$\frac{1}{k+1}\binom{n}{k} = \frac{1}{n+1}\binom{n+1}{k+1}$	k is trapped in a denominator (expected values)
Trinomial revision	$\binom{n}{m}\binom{m}{k} = \binom{n}{k}\binom{n-k}{m-k}$	"choose a set, then a subset" — re-associate
Upper negation	$\binom{-n}{k} = (-1)^k\binom{n+k-1}{k}$	turn an alternating sum into a positive one or back
Pascal	$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$	no closed form: build an $O(n^2)$ DP / telescope
Ratio	$\binom{n}{k+1} = \binom{n}{k}\frac{n-k}{k+1}$	walk along a row in $O(1)$ per step (only if you can divide)

Also: $\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1} = \frac{n}{n-k}\binom{n-1}{k}$, and $\binom{n}{k} = \binom{n-1}{k-1} \cdot \frac{n}{k}$ (useful in proofs; division-free forms are better in code).

Absorption in action: $\sum_k k\binom{n}{k}$

$\sum_k k\binom{n}{k} = \sum_k n\binom{n-1}{k-1} = n \cdot 2^{n-1}$. Each weighted sum below is "absorb, then a known sum".

Polynomial weights: use falling factorials, never powers

$k^2 = k^{\underline{2}} + k^{\underline{1}}$, $k^3 = k^{\underline{3}} + 3k^{\underline{2}} + k^{\underline{1}}$, $k^4 = k^{\underline{4}} + 6k^{\underline{3}} + 7k^{\underline{2}} + k^{\underline{1}}$ (coefficients are $S(m,j)$, §5). After rewriting, generalized absorption turns each $k^{\underline{j}}\binom{n}{k}$ into $n^{\underline{j}}\binom{n-j}{k-j}$.

3. Row, Column and Diagonal Sums of Pascal's Triangle

3.1 Full row

$$\sum_k \binom{n}{k} = 2^n, \qquad \sum_k (-1)^k \binom{n}{k} = [\,n = 0\,], \qquad \sum_{k \text{ even}} \binom{n}{k} = \sum_{k \text{ odd}} \binom{n}{k} = 2^{n-1} \ (n \ge 1).$$

$$\sum_k \binom{n}{k} x^k = (1+x)^n, \qquad \sum_k \binom{n}{k} x^k y^{n-k} = (x+y)^n, \qquad \sum_k \binom{n}{k} 2^k = 3^n.$$

3.2 Column (hockey stick) — upper index moves

$$\sum_{i=r}^{n}\binom{i}{r}=\binom{n+1}{r+1} \qquad \text{e.g. } \sum_{i\le n}\binom{i}{2}=\binom{n+1}{3}=\sum_{i<n} i \text{ pairs.}$$

3.3 Diagonal — both indices move

$$\sum_{i=0}^{n}\binom{r+i}{i}=\binom{r+n+1}{n} \qquad \text{(number of multisets: see stars and bars).}$$

3.4 Anti-diagonal

$$\sum_{k}\binom{n-k}{k}=F_{n+1} \quad (\text{Fibonacci}, F_1=F_2=1).$$

3.5 Partial row — NO closed form

$S(n,k)=\sum_{i=0}^{k}\binom{n}{i}$ has no closed form. It satisfies

$$S(n+1,k)=2S(n,k)-\binom{n}{k}, \qquad S(n,k+1)=S(n,k)+\binom{n}{k+1},$$

so (n,k) can move by ±1 in $O(1)$: answer many queries with **Mo's algorithm** in $O((N+q)\sqrt{q})$. Single query: just loop $O(k)$. Partial *alternating* rows do close: $\sum_{k=0}^{m}(-1)^k\binom{n}{k}=(-1)^m\binom{n-1}{m}$.

3.6 Parity and Pascal mod 2

$\binom{n}{k}$ is odd $\iff$ `(k & n) == k`. Number of odd entries in row n: $2^{\text{popcount}(n)}$. $v_2\binom{n}{k}=\text{popcount}(k)+\text{popcount}(n-k)-\text{popcount}(n)$.

4. Weighted Sums (polynomial in k)

Master results (via absorption):

$$\sum_{k}\binom{n}{k}k^{\underline{m}}=n^{\underline{m}}2^{n-m}, \qquad \sum_{k}\binom{n}{k}k^{m}=\sum_{j}S(m,j)\,n^{\underline{j}}2^{n-j}.$$

The second costs $O(m^2)$ independent of n, so $\sum_k\binom{n}{k}k^{100}$ for $n=10^{18}$ is cheap.

$\sum_k\binom{n}{k}k=n2^{n-1}$	$\sum_k\binom{n}{k}k^2=n(n+1)2^{n-2}$
$\sum_k\binom{n}{k}k^3=n^2(n+3)2^{n-3}$	$\sum_k\binom{n}{k}k(k-1)=n(n-1)2^{n-2}$
$\sum_k\binom{n}{k}kx^k=nx(1+x)^{n-1}$	$\sum_k\binom{n}{k}k^2x^k=nx(1+x)^{n-2}(1+nx)$
$\sum_k\frac{1}{k+1}\binom{n}{k}=\frac{2^{n+1}-1}{n+1}$	$\sum_k\frac{x^{k+1}}{k+1}\binom{n}{k}=\frac{(1+x)^{n+1}-1}{n+1}$
$\sum_k\frac{1}{(k+1)(k+2)}\binom{n}{k}=\frac{2^{n+2}-n-3}{(n+1)(n+2)}$	$\sum_k k\binom{n}{k}p^k(1-p)^{n-k}=np$ (binomial mean)

Variance of Binomial(n,p): $np(1-p)$. These two save you whenever "each of n items independently with probability p".

4.1 Power sums $\sum_{i=1}^{n} i^k$

$$\sum i = \frac{n(n+1)}{2}, \quad \sum i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum i^3 = \left(\frac{n(n+1)}{2}\right)^2, \quad \sum i^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}.$$

General k: falling-factorial route $\sum_{i=0}^{n} i^{\underline{j}} = \frac{(n+1)^{\underline{j+1}}}{j+1}$, so $\sum_{i\le n} i^k = \sum_j S(k,j)\frac{(n+1)^{\underline{j+1}}}{j+1}$ in $O(k^2)$. Or **Lagrange interpolation** in $O(k)$: $\sum_{i\le n} i^k$ is a polynomial of degree $k+1$ in n; evaluate at $0..k+1$ and interpolate.

```
// value at x of the polynomial with y[i] = P(i), i = 0..d   (needs fact/ifact, x may be huge)
long long lagrange(vector<long long>& y, long long x) {
    int d = y.size() - 1; x %= MOD;
    if (x ≤ d) return y[x];                                   // careful: only if original x ≤ d
    vector<long long> pre(d + 2, 1), suf(d + 2, 1);
    for (int i = 0; i ≤ d; i++) pre[i + 1] = pre[i] * ((x - i + MOD) % MOD) % MOD;
    for (int i = d; i ≥ 0; i--) suf[i] = suf[i + 1] * ((x - i + MOD) % MOD) % MOD;
    long long res = 0;
    for (int i = 0; i ≤ d; i++) {
        long long t = y[i] * pre[i] % MOD * suf[i + 1] % MOD * ifact[i] % MOD * ifact[d - i] % MOD;
        res = ((d - i) & 1) ? (res - t + MOD) % MOD : (res + t) % MOD;
    }
    return res;
}
```

Trap

Do the `x <= d` check on the *original* x, before reducing mod. If $x \ge \text{MOD}$, the reduced value may land in $[0, d]$ while the real x does not.

5. The Vandermonde Family (convolution shapes)

Signature: two binomials whose *lower* indices sum to a constant. Story: choose p people from m men and n women; split by how many men.

$$\boxed{\sum_k \binom{m}{k}\binom{n}{p-k} = \binom{m+n}{p}} \qquad \sum_k \binom{m}{k}\binom{n}{p+k} = \binom{m+n}{m+p} \ (n \ge 0).$$

$$\sum_k \binom{m}{k}\binom{n}{k} = \binom{m+n}{m}, \qquad \sum_k \binom{n}{k}^2 = \binom{2n}{n}, \qquad \sum_k \binom{n}{k}\binom{n}{k+m} = \binom{2n}{n+m}.$$

Weighted variants (absorb, then Vandermonde):

$$\sum_k k\binom{m}{k}\binom{n}{p-k} = m\binom{m+n-1}{p-1}, \qquad \sum_k k^{\underline{j}}\binom{m}{k}\binom{n}{p-k} = m^{\underline{j}}\binom{m+n-j}{p-j},$$

$$\sum_k k\binom{n}{k}^2 = n\binom{2n-1}{n-1}, \qquad \sum_k k^2\binom{n}{k}^2 = n^2\binom{2n-2}{n-1}.$$

Alternating: $\sum_k (-1)^k\binom{m}{k}\binom{n}{p-k} = [x^p](1-x)^m(1+x)^n$. Special: $\sum_k (-1)^k\binom{n}{k}^2 = (-1)^{n/2}\binom{n}{n/2}$ for even n, 0 for odd. Route: push $(-1)^k$ into one binomial by upper negation, then (V).

Multinomial Vandermonde: $\sum_{k_1+\cdots+k_r=p}\binom{n_1}{k_1}\cdots\binom{n_r}{k_r} = \binom{n_1+\cdots+n_r}{p}$. Triple products: apply (V) twice, innermost first.

The expansion direction — the move people miss

Read Vandermonde *right to left*: $\binom{a+b}{r} = \sum_k \binom{a}{k}\binom{b}{r-k}$. Breaking one binomial into a convolution can *create* a Vandermonde shape with another factor, after which you swap sums. Also: if your answer is $\sum_k f(k)g(p-k)$ with f, g arbitrary, it's a polynomial product — NTT in $O(n \log n)$.

6. Split a Line (generalized hockey stick)

The highest-value single formula for "sum over the position of a pivot":

$$\boxed{\sum_{k=0}^{l}\binom{l-k}{m}\binom{q+k}{n}=\binom{l+q+1}{m+n+1}}$$

Story: choose $m+n+1$ items from a line of $l+q+1$; the $(m+1)$-th chosen item is the pivot. Left of it you picked m, right of it n. Any sum "pick a split point, choose m on the left, n on the right" is one binomial.

$$\sum_{k=0}^{n}\binom{k}{a}\binom{n-k}{b}=\binom{n+1}{a+b+1},\qquad \sum_{i}\binom{a+i}{i}\binom{b+n-i}{n-i}=\binom{a+b+n+1}{n}.$$

Sum of the j-th smallest over all k-subsets of $\{1..n\}$

Element x is the j-th smallest in $\binom{x-1}{j-1}\binom{n-x}{k-j}$ subsets. So the answer is $\sum_x x\binom{x-1}{j-1}\binom{n-x}{k-j} = j\sum_x\binom{x}{j}\binom{n-x}{k-j} = j\binom{n+1}{k+1}$. Expected j-th smallest $= j\frac{n+1}{k+1}$.

7. Alternating Sums and Binomial Inversion

7.1 The vanishing theorem

$$\boxed{\sum_{k=0}^{n}(-1)^k\binom{n}{k}P(k)=0 \quad \text{whenever } \deg P<n}$$

If $\deg P=n$ with leading coefficient c, the sum is $(-1)^n c\,n!$. In general $\sum_k(-1)^k\binom{n}{k}f(k)=(-1)^n\Delta^n f(0)$. **Before doing any work on an alternating sum with a polynomial weight, check the degree.**

7.2 Specific alternating sums

$$\sum_k\frac{(-1)^k}{k+1}\binom{n}{k}=\frac{1}{n+1},\qquad \sum_k\frac{(-1)^k}{k+m}\binom{n}{k}=\frac{1}{m\binom{n+m}{m}},\qquad \sum_{k\ge1}\frac{(-1)^{k-1}}{k}\binom{n}{k}=H_n,$$

$$\sum_k(-1)^k\binom{n}{k}\binom{k}{m}=(-1)^m[n=m],\qquad \sum_{k=0}^{m}(-1)^k\binom{n}{k}=(-1)^m\binom{n-1}{m},$$

$$\sum_k(-1)^k\binom{n}{k}(n-k)^n=n!,\qquad \sum_k(-1)^k\binom{n}{k}(n-k)^m=0\ (m<n).$$

7.3 Binomial inversion ("at least" $\leftrightarrow$ "exactly")

$$f(n)=\sum_{k=0}^{n}\binom{n}{k}g(k)\iff g(n)=\sum_{k=0}^{n}(-1)^{n-k}\binom{n}{k}f(k),$$

$$f(k)=\sum_{i\ge k}\binom{i}{k}g(i)\iff g(k)=\sum_{i\ge k}(-1)^{i-k}\binom{i}{k}f(i).$$

How to use it

Let $g(i)$ = #ways where *exactly* the set of i "bad" positions is bad, and $f(k)$ = #ways after *forcing* some chosen k positions to be bad (others free), summed over the choice. Forcing is usually easy: $f(k)=\binom{n}{k}\cdot$ (count with k forced). Then exactly-0 bad $=\sum_k(-1)^k f(k)$; exactly-m bad $=\sum_{k\ge m}(-1)^{k-m}\binom{k}{m}f(k)$.

Derangements

Force k fixed points: $\binom{n}{k}(n-k)!$. So $D_n=\sum_k(-1)^k\binom{n}{k}(n-k)!=n!\sum_k\frac{(-1)^k}{k!}$. Permutations with exactly m fixed points: $\binom{n}{m}D_{n-m}$.

7.4 Inclusion–exclusion template

$$\left|\bigcap \overline{A_i}\right| = \sum_{S\subseteq[n]} (-1)^{|S|} \left|\bigcap_{i\in S} A_i\right| = \sum_k (-1)^k \binom{n}{k} a_k \quad \text{(when any } k \text{ sets intersect in } a_k\text{)}.$$

Examples: surjections $n \to k$: $\sum_j (-1)^j \binom{k}{j}(k-j)^n$. Colorings of a path using *all* k colors: $\sum_j (-1)^j \binom{k}{j}(k-j)(k-j-1)^{n-1}$.

8. Stars and Bars and Distributions

8.1 Core counts

Count	Answer
$x_1 + \cdots + x_k = n$, $x_i \geq 0$	$\binom{n+k-1}{k-1}$
$x_1 + \cdots + x_k = n$, $x_i \geq 1$	$\binom{n-1}{k-1}$
$x_i \geq a_i$	substitute $y_i = x_i - a_i$: $\binom{n-\sum a_i+k-1}{k-1}$
$x_1 + \cdots + x_k \leq n$, $x_i \geq 0$	$\binom{n+k}{k}$ (add a slack variable)
$0 \leq x_i \leq M$, sum $= n$	$\sum_j (-1)^j \binom{k}{j}\binom{n-j(M+1)+k-1}{k-1}$
$0 \leq x_i \leq M_i$ (different caps)	IE over subsets, or knapsack DP with prefix sums $O(nk)$
multisets of size n from k types	$\binom{n+k-1}{n}$
non-decreasing sequences of length n with values in $[1,k]$	$\binom{n+k-1}{n}$
strictly increasing, length n, values $[1,k]$	$\binom{k}{n}$
k-subsets of $\{1..n\}$, no two adjacent	$\binom{n-k+1}{k}$
same on a circle	$\frac{n}{n-k}\binom{n-k}{k}$
k-subsets with gaps $\geq d$ between chosen	$\binom{n-(k-1)(d-1)}{k}$

8.2 The twelvefold way (balls into boxes)

n balls, k boxes:

	any	≤ 1 **per box**	≥ 1 **per box**
labeled balls, labeled boxes	k^n	$k^{\underline{n}}$	$k!\,S(n,k)$
unlabeled balls, labeled boxes	$\binom{n+k-1}{n}$	$\binom{k}{n}$	$\binom{n-1}{k-1}$
labeled balls, unlabeled boxes	$\sum_{j\leq k} S(n,j)$	$[n \leq k]$	$S(n,k)$
unlabeled balls, unlabeled boxes	$p_{\leq k}(n)$ (partitions, $\leq k$ parts)	$[n \leq k]$	$p_k(n)$ (exactly k parts)

Partitions into exactly k parts: $p_k(n) = p_k(n-k) + p_{k-1}(n-1)$ ($O(nk)$ DP).

8.3 Multinomials

Arrangements of a word with letter counts $a_1, \ldots, a_r$: $\binom{n}{a_1,\ldots,a_r} = \frac{n!}{a_1!\cdots a_r!}$. $\binom{n}{a,b,c} = \binom{n}{a}\binom{n-a}{b}$. $\sum \binom{n}{a,b,c} x^a y^b z^c = (x+y+z)^n$; coefficient sum $= 3^n$.

9. Lattice Paths

Paths	Count
$(0,0) \to (a,b)$, steps R/U	$\binom{a+b}{a}$
through a point P	paths$(0 \to P)\cdot$paths$(P \to$ end$)$
avoiding k blocked cells	sort cells, $dp_i = \text{paths}(0 \to c_i) - \sum_{j<i} dp_j \cdot \text{paths}(c_j \to c_i)$; $O(k^2)$
$(0,0) \to (n,n)$ never above diagonal	C_n
$(0,0) \to (a,b)$, $a \geq b$, never above $y = x$	$\binom{a+b}{b} - \binom{a+b}{b-1} = \frac{a-b+1}{a+1}\binom{a+b}{b}$
never touching line $y = x + c$ (reflection)	$\binom{a+b}{a} - \binom{a+b}{a+c}$ (end must be valid)
strip $0 \leq y - x \leq w$ (two walls)	alternating reflections, sum over $\pm$ images: $O((a+b)/w)$ terms
± 1 walk of length n ending at h	$\binom{n}{(n+h)/2}$ (parity!)
Dyck-like with "up/flat/down" (Motzkin)	$M_n = \sum_k \binom{n}{2k} C_k$
with diagonal steps (Delannoy)	$D(m,n) = \sum_k \binom{m}{k}\binom{n}{k} 2^k$

Reflection principle in one line

Bad paths (touching the forbidden line L) are in bijection with *all* paths from the start reflected across L to the same end. So good = total − paths(reflected start → end). For the line $y = x + 1$ ("never above diagonal"), reflect $(0,0)$ to $(-1,1)$.

```
// paths from (0,0) to (H,W) avoiding blocked cells b[0..k-1] (sorted by (r,c)), last element = (H,W)
auto paths = [&](int r1, int c1, int r2, int c2) { return (r2 < r1 || c2 < c1) ? 0LL : C(r2 - r1 + c2 -
    c1, r2 - r1); };
for (int i = 0; i < (int)b.size(); i++) {
    dp[i] = paths(0, 0, b[i].first, b[i].second);
    for (int j = 0; j < i; j++) dp[i] = (dp[i] - dp[j] * paths(b[j].first, b[j].second, b[i].first, b[i
    ].second) % MOD + MOD) % MOD;
}
// answer = dp[last]
```

10. Named Families (recognize from the shape)

Object	Formula	Signature / counts
Catalan C_n	$\frac{1}{n+1}\binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1}$; $C_{n+1} = \sum_k C_k C_{n-k}$	balanced brackets, binary trees with n nodes, triangulations of $(n+2)$-gon, non-crossing matchings, stack-sortable perms
Ballot	$\frac{a-b}{a+b}\binom{a+b}{a}$	A always strictly ahead of B in count
m-ary trees	$\frac{1}{(m-1)n+1}\binom{mn}{n}$	cycle lemma / Lagrange
Narayana	$\frac{1}{n}\binom{n}{k}\binom{n}{k-1}$	Catalan refined by #peaks
Motzkin M_n	$\sum_k \binom{n}{2k} C_k$	paths with flat steps
Delannoy	$\sum_k \binom{m}{k}\binom{n}{k} 2^k$	grid paths with diagonals
Schröder	$\sum_k \binom{n}{k}\binom{n+k}{k}\frac{1}{k+1}$	diagonal paths below a line
Derangements	$D_n = (n-1)(D_{n-1}+D_{n-2}) = nD_{n-1}+(-1)^n$	no fixed points; $D_n \approx n!/e$
Surjections	$k!S(n,k) = \sum_j (-1)^j \binom{k}{j}(k-j)^n$	every box nonempty
Stirling 2nd	$S(n,k) = kS(n-1,k) + S(n-1,k-1)$	set partitions into k blocks
Stirling 1st	$c(n,k) = (n-1)c(n-1,k) + c(n-1,k-1)$	perms with k cycles; $\sum_k c(n,k)x^k = x^{\overline{n}}$
Bell B_n	$B_{n+1} = \sum_k \binom{n}{k} B_k$	all set partitions: $1, 1, 2, 5, 15, 52, 203$
Eulerian	$A(n,k) = (k+1)A(n-1,k) + (n-k)A(n-1,k-1)$	perms with k descents
Partitions $p(n)$	pentagonal recurrence $O(n\sqrt{n})$	unordered sums
Fibonacci	$F_{n+1} = \sum_k \binom{n-k}{k}$	tilings by 1×1, 1×2
Labeled trees	n^{n-2} (Cayley); forests with k given roots: kn^{n-k-1}	spanning trees of K_n
Involutions	$I_n = I_{n-1} + (n-1)I_{n-2}$	perms with $p^2 = \mathrm{id}$

Conversions: $x^n = \sum_k S(n,k)x^{\underline{k}}$, $x^{\underline{n}} = \sum_k (-1)^{n-k} c(n,k) x^k$, $x^n = \sum_k A(n,k)\binom{x+k}{n}$ (Worpitzky).

```
// Stirling 2nd kind table, O(n^2)
S[0][0] = 1;
for (int i = 1; i ≤ n; i++) for (int k = 1; k ≤ i; k++)
    S[i][k] = (S[i - 1][k - 1] + (long long)k * S[i - 1][k]) % MOD;
// single row S(n, k) for fixed n, all k: O(k log) via k! S(n,k) = sum (-1)^j C(k,j) (k-j)^n  (or NTT
    for all k)
```

11. Roots of Unity Filter

To keep only the terms with $k \equiv r \pmod m$:

$$\sum_{k\equiv r\,(m)} \binom{n}{k} = \frac{1}{m}\sum_{j=0}^{m-1} \omega^{-jr}(1+\omega^j)^n, \qquad \omega = e^{2\pi i/m}.$$

For any $f(x) = \sum a_k x^k$: $\sum_{k\equiv r} a_k = \frac{1}{m}\sum_j \omega^{-jr} f(\omega^j)$.

- $m = 2$ (no roots needed): $\sum_{k \text{ even}} a_k = \frac{f(1)+f(-1)}{2}$, odd $= \frac{f(1)-f(-1)}{2}$.
- Mod p: an m-th root of unity exists iff $m \mid p-1$: $\omega = g^{(p-1)/m}$ (g = primitive root; 3 for 998244353, where $p-1 = 2^{23}\cdot 7\cdot 17$).
- $m \nmid p-1$: work in $\mathbb{F}_p[x]/(x^m-1)$ — keep length-m vectors, multiply cyclically, exponentiate by squaring. $O(m^2 \log n)$.
- $\sum_{k\equiv 0\,(3)} \binom{n}{k} = \frac{2^n + 2\cos(n\pi/3)}{3}$.

12. Generating Function Dictionary

Fallback when nothing matches: write your answer as $[x^k]$ of a product, simplify, read off.

Series	$[x^k]$	Series	$[x^k]$
$(1+x)^n$	$\binom{n}{k}$	$\frac{1-\sqrt{1-4x}}{2x}$	C_k
$\frac{1}{(1-x)^n}$	$\binom{n+k-1}{k}$	$\frac{1}{\sqrt{1-4x}}$	$\binom{2k}{k}$
$\frac{1}{1-x}$	1 (multiply = prefix sum)	$\frac{1}{\sqrt{1-4x}}\left(\frac{1-\sqrt{1-4x}}{2x}\right)^a$	$\binom{2k+a}{k}$
$\frac{x^a}{(1-x)^{b+1}}$	$\binom{k-a+b}{b}$	$\frac{x}{1-x-x^2}$	F_k
$\frac{1-x^{c+1}}{1-x}$	$[0 \le k \le c]$ (capped var)	$\frac{1}{1-ax}$	a^k
e^x (EGF)	$1/k!$	$\frac{(e^x-1)^m}{m!}$ (EGF)	$S(k,m)/k!$
$\frac{e^{-x}}{1-x}$ (EGF)	$D_k/k!$	$\exp(e^x-1)$ (EGF)	$B_k/k!$

OGF vs EGF: unlabeled pieces glued $\to$ multiply OGFs. Labeled pieces (the product counts relabelings) $\to$ multiply EGFs, i.e. $c_n = \sum_k \binom{n}{k} a_k b_{n-k}$.

Grid extractors: $[x^n y^m]\frac{1}{1-x-y} = \binom{n+m}{n}$, $[x^n y^m]\frac{1}{1-x-y-xy} = D(n,m)$.

Lagrange inversion (if $A(x) = x\varphi(A(x))$): $[x^n]A^k = \frac{k}{n}[u^{n-k}]\varphi(u)^n$.

Snake oil: for $\sum_k a_{n,k}$, form $F(x) = \sum_n (\text{sum}) x^n$, swap sums so the inner one is over n, recognize that GF, read the coefficient back.

13. Convolution Identities Worth Recognizing

$$\sum_k \binom{2k}{k}\binom{2n-2k}{n-k} = 4^n, \quad \sum_k C_k C_{n-k} = C_{n+1}, \quad \sum_k \binom{n}{k} F_k = F_{2n}, \quad \sum_k \binom{n}{k}\binom{n}{k+m} = \binom{2n}{n+m},$$

$$\sum_k \binom{n}{k} a^k b^{n-k} = (a+b)^n, \quad \sum_k \binom{m}{k}\binom{n+k}{p} \to \text{expand } \binom{n+k}{p} \text{ by Vandermonde first, then swap.}$$

$$\sum_k \binom{n}{k}\binom{k}{j} = \binom{n}{j} 2^{n-j}, \qquad \sum_k \binom{n}{k}\binom{k}{j}(-1)^k = (-1)^n [\, n = j \,].$$

14. Contribution Technique Formulas (sums over subsets)

These are the "sum over all subsets / subsequences" shapes that appear constantly. Sort the array $a_1 \le \cdots \le a_n$.

Quantity	Formula
$\sum_S \max(S)$ over nonempty subsets	$\sum_i a_i 2^{i-1}$
$\sum_S \min(S)$	$\sum_i a_i 2^{n-i}$
$\sum_{\lvert S\rvert=k} \max(S)$	$\sum_i a_i \binom{i-1}{k-1}$
$\sum_{\lvert S\rvert=k} \min(S)$	$\sum_i a_i \binom{n-i}{k-1}$
$\sum_S (\max - \min)$	$\sum_i a_i (2^{i-1} - 2^{n-i})$
$\sum_S \sum_{x\in S} x$ over all subsets	$2^{n-1} \sum a_i$
$\sum_{\lvert S\rvert=k} \sum_{x\in S} x$	$\binom{n-1}{k-1} \sum a_i$
$\sum_{\text{subarrays}} \sum$	$\sum_i a_i \cdot i(n-i+1)$
$\sum_{i<j} (a_j - a_i)$ (sorted)	$\sum_i a_i (2i - n - 1)$
#subsequences of s equal to t	DP $dp[j] \mathrel{+}= dp[j-1]$ when $s_i = t_j$
k-th order statistic of random k-subset	$\mathbb{E}[j\text{-th smallest}] = j\frac{n+1}{k+1}$

The contribution habit

Instead of iterating over subsets, ask for each element: "in how many subsets is this element the max / counted / the pivot?" That count is almost always a single binomial or a power of 2.

15. Probability and Expectation Shortcuts

- **Linearity:** $\mathbb{E}[\sum X_i] = \sum \mathbb{E}[X_i]$ even when dependent. Write the count as a sum of indicators.
- $\mathbb{E}[\text{fixed points of random perm}] = 1$; $\mathbb{E}[\#\text{cycles}] = H_n$; $\mathbb{E}[\#\text{records/left-to-right maxima}] = H_n$; $\mathbb{E}[\#\text{inversions}] = \frac{n(n-1)}{4}$.
- **Coupon collector:** $\mathbb{E} = nH_n$. Geometric with success p: $\mathbb{E} = 1/p$.
- $P(\text{exactly } k \text{ successes}) = \binom{n}{k} p^k (1-p)^{n-k}$.
- $\mathbb{E}[X] = \sum_{t \geq 1} P(X \geq t)$ for integer $X \geq 0$; $\mathbb{E}[\max]$ via $P(\max \leq t) = \prod P(X_i \leq t)$.
- Expected value mod P: compute numerator $\times$ inverse of denominator.
- Hypergeometric (draw k of N, K good): $P(j \text{ good}) = \binom{K}{j}\binom{N-K}{k-j}/\binom{N}{k}$; mean kK/N.

16. NT $\times$ Combinatorics Crossovers

- **Lucas:** $\binom{n}{r} \equiv \prod_i \binom{n_i}{r_i} \pmod p$ (base-p digits). Zero iff some $r_i > n_i$.
- **Kummer:** $v_p\binom{n}{r}$ = number of carries adding $r + (n-r)$ in base p. **Legendre:** $v_p(n!) = \sum_k \lfloor n/p^k \rfloor = \frac{n - s_p(n)}{p-1}$.
- $p \mid \binom{p}{k}$ for $0 < k < p$; $\binom{p-1}{k} \equiv (-1)^k$; $\binom{2p}{p} \equiv 2 \pmod{p^3}$ for $p \geq 5$.
- $n \mid \binom{n}{k} \cdot \gcd(n,k)$; gcd of the row $\binom{n}{1}..\binom{n}{n-1}$ is p if $n = p^k$, else 1.
- Necklaces: $\frac{1}{n}\sum_{d|n} \varphi(n/d)k^d$; with exactly m ones: $\frac{1}{n}\sum_{d|\gcd(n,m)} \varphi(d)\binom{n/d}{m/d}$.
- r-tuples in $[1, M]^r$ with gcd 1: $\sum_d \mu(d)\lfloor M/d \rfloor^r$.
- Division by $k!$ when $k \geq P$: impossible — use Lucas or keep p-adic valuations separately.

17. Rarer but Occasionally Decisive

$$\sum_k \binom{n}{k} x(x+k)^{k-1}(y+n-k)^{n-k} = (x+y+n)^n \text{ (Abel)}, \qquad \sum_k (-1)^k \binom{2n}{n+k}^3 = \frac{(3n)!}{(n!)^3} \text{ (Dixon)},$$

$\sum_k \binom{n}{k}^3$ (Franel): no closed form; $(n+1)^2 F_{n+1} = (7n^2+7n+2)F_n + 8n^2 F_{n-1}$, $\qquad \sum_k \binom{n}{k}\binom{n+k}{k}(-1)^k = (-1)^n$.

No closed form? The sum is probably P-recursive (linear recurrence with polynomial coefficients). Brute-force ~ 30 terms, guess the recurrence (Gaussian elimination / Berlekamp–Massey for constant coefficients), verify, then run it in $O(n)$.

18. How to Actually Use This

1. Write the brute force as nested sums. Don't be clever yet.
2. Swap the order of summation. Inner sum over the index that appears in the most places.
3. Kill loose polynomial factors with absorption (falling factorials first; downward absorption for $\frac{1}{k+1}$).
4. Look at the indices:
   - lower indices sum to a constant $\to$ Vandermonde (§5);
   - one counts up, the other down / upper indices sum to a constant $\to$ split-a-line (§6);
   - single index counting up a column $\to$ hockey stick (§3).
5. Alternating? Check the polynomial degree first — it may be 0 (§7).
6. "At least" is easy but you need "exactly" $\to$ binomial inversion / IE (§7.3).
7. Still nothing: generating functions (§12), or brute force $n \leq 8$ and search OEIS / guess a recurrence.

Verification discipline

Every closed form gets checked against brute force on all small inputs. Boundary cases ($k > n$, $n = 0$, empty sums, negative indices passed to `C`) are where identities silently break. A 60-second comparison loop catches all of them.

19. Recognize the Problem → Jump Here

You see...	**Go to**
need $\binom{n}{k}$ with huge n / small mod / composite mod	§1
$\sum \binom{n}{k} \cdot$ (poly in k)	§4
$\sum_k \binom{\cdot}{k}\binom{\cdot}{p-k}$	§5 Vandermonde
"pivot position", j-th smallest, split a sequence	§6 split a line
alternating sum, "exactly k bad"	§7
distribute items, bounded variables, no two adjacent	§8
grid paths, obstacles, never cross a line	§9
brackets, trees, triangulations, partitions	§10 named families
only indices $\equiv r \pmod m$	§11 roots of unity
sum of max/min over subsets	§14 contribution
expected value	§15
$\binom{n}{k} \bmod p$ with n huge, parity of $\binom{n}{k}$	§16 Lucas/Kummer, §3.6